

1. (a) Find all the critical points of each of the functions below:

(i) $f(x, y) = \arctan(x + y) - y^2 - x$;

We seek (x, y) such that

$$\begin{aligned} \begin{cases} f_x(x, y) = 0 \\ f_y(x, y) = 0 \end{cases} &\Leftrightarrow \begin{cases} \frac{1}{1 + (x + y)^2} - 1 = 0 \\ \frac{1}{1 + (x + y)^2} - 2y = 0 \end{cases} \Leftrightarrow \begin{cases} \frac{1}{1 + (x + y)^2} = 1 \\ \frac{1}{1 + (x + y)^2} = 2y \end{cases} \\ &\Rightarrow \begin{cases} 2y = 1 \\ x + y = 0 \end{cases} \Rightarrow \boxed{(x, y) = \left(-\frac{1}{2}, \frac{1}{2}\right)}. \end{aligned}$$

(ii) $g(x, y) = \arccos(x - y) + 2x - y^2$.

We seek (x, y) such that

$$\begin{aligned} \begin{cases} g_x(x, y) = 0 \\ g_y(x, y) = 0 \end{cases} &\Leftrightarrow \begin{cases} \frac{-1}{\sqrt{1 - (x - y)^2}} + 2 = 0 \\ \frac{1}{\sqrt{1 - (x - y)^2}} - 2y = 0 \end{cases} \Leftrightarrow \begin{cases} \frac{1}{\sqrt{1 - (x - y)^2}} = 2, \\ \frac{1}{\sqrt{1 - (x - y)^2}} = 2y. \end{cases} \\ &\Rightarrow \begin{cases} y = 1 \\ x - 1 = \pm \frac{\sqrt{3}}{2} \end{cases} \Rightarrow \boxed{(x, y) = \left(1 - \frac{\sqrt{3}}{2}, 1\right), \left(1 + \frac{\sqrt{3}}{2}, 1\right)}. \end{aligned}$$

(iii) $h(x, y) = 2 \sin(x - 2y) + \cos(3y)$;

We seek (x, y) such that

$$\begin{aligned} \begin{cases} h_x(x, y) = 0 \\ h_y(x, y) = 0 \end{cases} &\Leftrightarrow \begin{cases} 2 \cos(x - 2y) = 0 \\ -4 \cos(x - 2y) - 3 \sin(3y) = 0 \end{cases} \Leftrightarrow \begin{cases} \cos(x - 2y) = 0, \\ \sin(3y) = 0. \end{cases} \\ &\Rightarrow \begin{cases} x - 2y = \frac{\pi}{2} + k\pi, & k \in \mathbb{Z}, \\ 3y = n\pi, & n \in \mathbb{Z} \end{cases} \quad (*) \\ &\Rightarrow \boxed{(x, y) = \left(\frac{\pi}{2} + \frac{2n\pi}{3} + k\pi, \frac{n\pi}{3}\right), \quad k, n \in \mathbb{Z}}. \end{aligned}$$

- (b) Select all the critical points of the function h lying in the region

$$D := \{(x, y) \in \mathbb{R}^2: x - 2y > 0 \text{ and } y > 0\}.$$

$$(x, y) \in D \Leftrightarrow \begin{cases} x - 2y > 0, \\ y > 0 \end{cases} \stackrel{(*)}{\Leftrightarrow} k \in \mathbb{N} \text{ and } n \in \mathbb{N} \setminus \{0\}.$$

So, the critical points of the function h lying in the region D are

$$(x, y) = \left(\frac{\pi}{2} + \frac{2n\pi}{3} + k\pi, \frac{n\pi}{3} \right), \quad k \in \mathbb{N} \text{ and } n \in \mathbb{N} \setminus \{0\}.$$

2. Determine the nature (points of local maximum/minimum, saddle point) of each critical point of the function:

(a) $f(x, y) := 25x^3 + 12y^3 - 75x^2 - 36y;$

We seek (x, y) such that

$$\begin{cases} f_x(x, y) = 0 \\ f_y(x, y) = 0 \end{cases} \Leftrightarrow \begin{cases} 75x^2 - 150x = 0 \\ 36y^2 - 36 = 0 \end{cases} \Leftrightarrow (x, y) = (0, -1), (0, 1), (2, -1), (2, 1).$$

$$f_{xx} = 150x - 150, \quad f_{xy} = 0, \quad f_{yy} = 72y.$$

$$D(0, -1) = \begin{vmatrix} -150 & 0 \\ 0 & -72 \end{vmatrix} = 10800 > 0 \text{ and } f_{xx}(0, -1) < 0.$$

So, $(0, -1)$ is a point of local maximum.

$$D(0, 1) = \begin{vmatrix} -150 & 0 \\ 0 & 72 \end{vmatrix} = -10800 < 0 \Rightarrow (0, 1) \text{ is a saddle point.}$$

$$D(2, -1) = \begin{vmatrix} 150 & 0 \\ 0 & -72 \end{vmatrix} = -10800 < 0 \Rightarrow (2, -1) \text{ is a saddle point.}$$

$$D(2, 1) = \begin{vmatrix} 150 & 0 \\ 0 & 72 \end{vmatrix} = 10800 > 0 \text{ and } f_{xx}(2, 1) > 0$$

Hence, $(2, 1)$ is a point of local minimum.

(b) $f(x, y) = 2x^2 - y^3 - 2xy;$

$\nabla f(x, y) = (4x - 2y, -3y^2 - 2x)$. So the critical points of f are the points (x, y) such that

$$\begin{cases} 4x - 2y = 0, \\ -3y^2 - 2x = 0 \end{cases} \Leftrightarrow (x, y) = (0, 0), \left(-\frac{1}{6}, -\frac{1}{3}\right).$$

Now

$$f_{xx} = 4, \quad f_{xy} = -2, \quad f_{yy} = -6y$$

$$D(0, 0) = \begin{vmatrix} f_{xx}(0, 0) & f_{xy}(0, 0) \\ f_{xy}(0, 0) & f_{yy}(0, 0) \end{vmatrix} = \begin{vmatrix} 4 & -2 \\ -2 & 0 \end{vmatrix} = -4 < 0 \Rightarrow \text{saddle point at } (0, 0).$$

$$D\left(-\frac{1}{6}, -\frac{1}{3}\right) = \begin{vmatrix} f_{xx}\left(-\frac{1}{6}, -\frac{1}{3}\right) & f_{xy}\left(-\frac{1}{6}, -\frac{1}{3}\right) \\ f_{xy}\left(-\frac{1}{6}, -\frac{1}{3}\right) & f_{yy}\left(-\frac{1}{6}, -\frac{1}{3}\right) \end{vmatrix} = \begin{vmatrix} 4 & -2 \\ -2 & 2 \end{vmatrix} = 4 > 0 \text{ and } f_{xx}\left(-\frac{1}{6}, -\frac{1}{3}\right) = 4 > 0$$

$\left(-\frac{1}{6}, -\frac{1}{3}\right)$ is a point of local minimum.

(c) $f(x, y) = x^2 - \frac{4xy}{y^2 + 1};$

$\nabla f(x, y) = \left(2x - \frac{4y}{y^2 + 1}, -\frac{4x(1 - y^2)}{(y^2 + 1)^2} \right)$. So the critical points of f are the points (x, y) such that

$$\begin{cases} 2x - \frac{4y}{y^2 + 1} = 0, \\ -\frac{4x(1 - y^2)}{(y^2 + 1)^2} = 0 \end{cases} \Leftrightarrow (x, y) = (0, 0), (-1, -1), (1, 1).$$

Now,

$$f_{xx} = 2, \quad f_{xy} = -\frac{4(1 - y^2)}{(y^2 + 1)^2}, \quad f_{yy} = -\frac{8xy(3y^2 - 1)}{(y^2 + 1)^3}.$$

$$D(0, 0) = \begin{vmatrix} f_{xx}(0, 0) & f_{xy}(0, 0) \\ f_{xy}(0, 0) & f_{yy}(0, 0) \end{vmatrix} = \begin{vmatrix} 2 & -4 \\ -4 & 0 \end{vmatrix} = -16 < 0 \Rightarrow \text{saddle point at } (0, 0).$$

$$D(-1, -1) = \begin{vmatrix} f_{xx}(-1, -1) & f_{xy}(-1, -1) \\ f_{xy}(-1, -1) & f_{yy}(-1, -1) \end{vmatrix} = \begin{vmatrix} 4 & 0 \\ 0 & -2 \end{vmatrix} = -4 < 0 \Rightarrow \text{saddle point at } (-1, -1)$$

Also $D(1, 1) = -4 < 0$ and hence, $(1, 1)$ is also a saddle point.

(d) $f(x, y) = e^{2x - x^2 - y^2};$

$\nabla f(x, y) = (2 - 2x, -2y)e^{2x - x^2 - y^2}$. Hence $\nabla f(x, y) = (0, 0) \Rightarrow x = 1$ and $y = 0$. So the only critical point is $(1, 0)$. To find the nature of this critical point we compute the second order partial derivative

$$f_{xx} = [-2 + (2 - 2x)^2]e^{2x - x^2 - y^2}, \quad f_{xy} = -2y(2 - 2x)e^{2x - x^2 - y^2}, \quad f_{yy} = (-2 + 4y^2)e^{2x - x^2 - y^2}.$$

So, $f_{xx}(1, 0) = -2e$, $f_{xy}(1, 0) = 0$, $f_{yy}(1, 0) = -2e$. Hence,

$$D(1, 0) = \begin{vmatrix} -2e & 0 \\ 0 & -2e \end{vmatrix} = 4e^2 > 0 \quad \text{and} \quad f_{xx}(1, 0) = -2e < 0 \Rightarrow$$

f has a local maximum at $(1, 0)$.

3. Consider the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by

$$f(x, y) = \int_{-x^2}^{2y} \cos(t^2) dt \quad \forall (x, y) \in \mathbb{R}^2.$$

Find **All** the critical points of the function f .

A point $(x, y) \in \mathbb{R}^2$ is a critical point to the function f if

$$\begin{aligned} \begin{cases} f_x(x, y) = 0 \\ f_y(x, y) = 0 \end{cases} &\Rightarrow \begin{cases} 2x \cos(x^4) = 0 \\ 2 \cos(4y^2) = 0 \end{cases} \Rightarrow \begin{cases} x = 0 \text{ or } \cos(x^4) = 0 \\ 2 \cos(4y^2) = 0 \end{cases} \\ &\Rightarrow \begin{cases} x = 0 \text{ or } x = \pm \left(\frac{\pi}{2} + n\pi \right)^{\frac{1}{4}}, \quad n \in \mathbb{N} \text{ (Explain why!)} \\ y = \pm \frac{1}{2} \left(\frac{\pi}{2} + m\pi \right)^{\frac{1}{2}}, \quad m \in \mathbb{N} \text{ (Explain why!)} \end{cases} \end{aligned}$$

So, the critical points of the function f are:

$$\begin{aligned} & \left(0, -\frac{1}{2}\left(\frac{\pi}{2} + m\pi\right)^{\frac{1}{2}}\right), & \left(0, \frac{1}{2}\left(\frac{\pi}{2} + m\pi\right)^{\frac{1}{2}}\right), \\ & \left(-\left(\frac{\pi}{2} + n\pi\right)^{\frac{1}{4}}, -\frac{1}{2}\left(\frac{\pi}{2} + m\pi\right)^{\frac{1}{2}}\right), & \left(-\left(\frac{\pi}{2} + n\pi\right)^{\frac{1}{4}}, \frac{1}{2}\left(\frac{\pi}{2} + m\pi\right)^{\frac{1}{2}}\right) \quad m, n \in \mathbb{N}, \\ & \left(\left(\frac{\pi}{2} + n\pi\right)^{\frac{1}{4}}, -\frac{1}{2}\left(\frac{\pi}{2} + m\pi\right)^{\frac{1}{2}}\right), & \left(\left(\frac{\pi}{2} + n\pi\right)^{1/4}, \frac{1}{2}\left(\frac{\pi}{2} + m\pi\right)^{\frac{1}{2}}\right), \end{aligned}$$